Data Science in the Wild – Group Project

Project Title: LaunchPad Nexus

Client: Newcastle University Careers & Employability Service

Company: LaunchPad Nexus

Project Purpose: LaunchPad Nexus is a student-built AI-supported tool that helps final-year and recent graduates targeting data and technology roles turn noisy job descriptions into clear, personalised 7-day action plans. Students share a brief profile and upload or paste a real job description; the system filters out ambiguous or unrealistic requirements, highlights what genuinely matters, and generates a 3-step weekly action plan. A built-in feedback loop captures which actions students complete and how these affect their confidence, enabling continuous improvement of the model and the support it provides.

Team Members:

Pranit Chatterjee
Xiaokai Chen
Ajay Sivakumar
Vivan Kushal Heneger
Archie Barnett
Rajbala Jadhav

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Executive Report on LaunchPad Nexus

Executive Summary

In the highly competitive global graduate environment, universities face increasing pressure to support learners in achieving strong employment outcomes. While many students access careers services, job boards and employability events, these supports are often fragmented, generic and reactive. As a result, domestic final-year students and recent graduates feel overwhelmed by the job search process, unsure which actions will most improve their chances of success, and likely to waste time on low-impact applications.

This report proposes LaunchPad Nexus, an AI-supported web application that helps final-year students and recent graduates targeting data and technology roles turn noisy job descriptions into clear, personalised 7-day action plans. Students share a brief profile and upload or paste a real job description; the system filters out ambiguous or unrealistic requirements, highlights what genuinely matters, and generates a 3–5 step weekly action plan. A feedback loop captures which actions students complete and how this affects their confidence, enabling iterative improvement of the model and the support it provides.

The proposed minimum viable product (MVP) can be piloted within eight weeks using existing digital infrastructure and limited staff support. The main resources required are a web-based front end, LLM-enabled recommendation logic, a curated database of graduate case studies, and oversight from careers staff. The expected benefits include improved job-search clarity for students, more efficient use of support services, stronger graduate outcomes, and a clearer evidence base for future service improvement.

The decision requested is approval to pilot LaunchPad Nexus with a defined cohort of domestic students, evaluate engagement and impact, and refine the tool for wider adoption if successful.

Introduction (Context)

The original challenge addressed by this project is: In the highly competitive global graduate environment, how can we support learners in securing their preferred outcome? For this report, our group refined the challenge by focusing on a specific customer segment: domestic UK final-year students and recent graduates seeking employment in data and technology roles. This narrowing of scope was deliberate. The broader challenge encompasses many different learner goals, including further study, entrepreneurship and international mobility, each with distinct needs. By focusing on employment-seeking domestic students, we were able to define the problem more precisely and design a more realistic and testable MVP.

Problem Definition

Most available guidance is generic, fragmented, or inaccessible at the moment students need it most (AGCAS, 2023). Job descriptions in data and technology roles frequently contain long lists of requirements, many of which are aspirational, duplicated, or less important than they appear. Students spend significant time interpreting these requirements, often unsure which to prioritise and which to disregard. This leads to unfocused preparation, low-confidence applications, and poor response rates. The council — represented here by the university's Careers and Employability Service — lacks a scalable tool to provide timely, personalised and actionable guidance at the moment of need (HESA, 2024). The relationship between higher education quality and graduate labour market outcomes is well-established in the UK context (Smith, McKnight and Naylor, 2000; Wadsworth, 2010). Graduate employability research has consistently shown that students struggle to translate academic achievement into successful job applications, particularly in competitive technical fields (Crawford and Fink, 2019–2020; Wilton, 2011).

Organisational Goals

To address these issues, the university and its Careers Service seek to:

1. Improve graduate employment outcomes and satisfaction scores.
2. Provide personalised employability support at scale without proportionally increasing staff cost.
3. Generate behavioural data on student job-search patterns to inform future service design.
4. Build a reusable digital tool that can be extended or white-labelled to other institutions.

Key Stakeholders Involved

1. Internal: Newcastle University Careers & Employability Service, academic schools, student success teams.
2. External:
   - Students – primary users and source of behavioural data.
   - Graduate employers – contribute case studies and validate the relevance of requirements.
   - LLM API providers (e.g. Google Gemini / AI Studio) – supply the intelligence layer.
   - Alumni network – optional future contributors of career pathway data.

Previous Attempts & Their Challenges

Previous attempts to address this problem include careers workshops, CV clinics, job fairs, one-to-one appointments, and generic online resources. While valuable, these interventions share common limitations: they are often one-size-fits-all, dependent on students self-selecting into support, and not available at the moment a student is reviewing a specific job description (Universities UK, 2023). More recent digital tools, including chatbots and AI-enhanced guidance platforms, have emerged but most remain at an early stage and lack personalisation or feedback loops (Prospects, 2023). Our analysis concluded that the gap is not an absence of support, but a lack of personalised, timely and action-focused guidance at the point of need.

Problem Analysis (Need)

The customer segment selected is domestic final-year students and recent graduates seeking employment in data and technology roles. Through our customer journey analysis (see Appendix B), we identified that these students often experience the job search as fragmented, stressful and difficult to navigate. They may spend significant time browsing multiple job platforms, submitting large numbers of applications, and trying to interpret conflicting advice, yet still receive few interviews or offers.

Customer Journey & Key Pain Points

The Customer Journey Map (Figure 2, Appendix B) identifies six key stages: Awareness; Exploration; Application Preparation; Submission; Waiting and Response; and Outcome. Key pain points are concentrated in the Exploration and Application Preparation stages, where students lack personalised guidance on which requirements to prioritise. The Empathy Map (Figure 1, Appendix A) further highlights four pain clusters:

- Overwhelm from complex, lengthy job descriptions with ambiguous or excessive requirements.
- Inability to prioritise — students do not know which actions will most improve their success.
- Lack of feedback and accountability, making it impossible to know whether preparation is working.
- Confidence gap — particularly for students from non-traditional or international backgrounds interpreting UK employer expectations.

A common pattern across the journey is that students are busy, but not always progressing in the most effective direction. They need not more information, but the right next action at the right time.

This need is supported by prior research. Shury et al. (2017) found that having a clearer career plan was one of the most important factors associated with positive graduate outcomes, while a focused approach to job applications was linked to better employment results. The Institute of Student Employers (ISE, 2023) noted that graduates who engaged in structured job-search preparation were significantly more likely to receive offers within three months of graduation. This is consistent with Robinson, Garton and Vaughn's (2007) finding that graduates benefit most from preparation that is specific, skill-referenced and tied directly to employer expectations.

Impact on the Organisation & Stakeholders

When students lack clear direction, demand for careers support becomes reactive and uneven, with some students seeking urgent help late in the recruitment cycle while others disengage entirely. This makes it difficult for careers teams to deliver support efficiently at scale. Lower graduate employment outcomes negatively affect institutional performance indicators, league table positions, and the perceived value of the degree (HESA, 2024). Employers also face consequences indirectly, receiving less well-prepared applicants (CBI, 2022).

Key Findings from Our Research

Mapping the customer journey showed that the most significant issue was not simply finding jobs, but helping students identify the right next action at the right time. This insight reshaped our solution. Rather than creating another general careers information platform, we focused on building an action-prioritisation tool that reduces overwhelm by analysing a specific job description against a student's specific profile, then generating a concrete weekly plan.

Proposed Solution (Vision)

Our proposed solution is LaunchPad Nexus, a web-based AI-supported employability application. The application asks the user to provide structured profile information, including degree subject, career goal, skills, experience level, application history, confidence level and available time. The student then uploads or pastes a real job description. The system uses a large language model (LLM), informed by curated graduate employment case studies, to analyse the job description and the student's profile together.

How LaunchPad Nexus Works

The core process has five stages:

1. Profile & Intake: Collect goals, skills, timeline and constraints via a structured form.
2. JD Noise Filtering: The LLM analyses the job description, removing aspirational or duplicated requirements and surfacing what genuinely matters for this role at this level.
3. Gap Diagnosis: The system compares the filtered requirements against the student's profile to identify the key gaps.
4. Action Plan Generation: The system recommends 3–5 priority next steps for the coming 7 days. Actions may include tailoring a CV, building a portfolio project, practising specific interview questions, or improving a LinkedIn profile.
5. Nudge, Track & Learn: Students track completed actions and provide feedback. The system uses this data to refine future recommendations.

The Value Proposition Canvas (Appendix C, Figures 2 and 3) illustrates precisely how the solution relieves the key pains (overwhelm, lack of direction, no feedback) and creates the desired gains (clarity, confidence, targeted progress) for the target customer.

How Our Solution Solves Challenges

Challenge	Solution
Overwhelming job descriptions	LLM noise-filtering extracts what genuinely matters for this role
No personalised guidance at scale	AI-generated weekly action plan matched to student profile and JD
Lack of feedback / accountability	Built-in action tracking and post-use feedback loop
Generic careers support	Case-study-informed LLM gives role-specific, level-appropriate advice
Students engage too late	On-demand tool usable at the exact moment of reviewing a job listing

Table 1: Key challenges and the LaunchPad Nexus solution for each. Source: Authors' own, based on student interviews and customer journey mapping.

Critical Assumptions

1. Students are willing to input accurate and sufficient profile information into the tool.
2. LLM-based recommendations informed by curated case studies will be relevant and actionable enough to influence student behaviour.
3. Students will respond positively to structured weekly guidance and return to update their progress.
4. The MVP can be integrated into the university's employability support ecosystem without large-scale infrastructure changes.

Outcomes (Expected Results)

The minimum viable product for LaunchPad Nexus is a functioning web application prototype that accepts student input, processes that input through an LLM-supported recommendation workflow, and returns a personalised weekly action plan. The MVP's purpose is to test whether students find the recommendations useful, understandable and motivating, and whether the tool helps them make more effective progress in their job search.

MVP Deployment and Ownership

When the MVP is deployed, students in the pilot cohort will have access to a structured, personalised employability support tool available on demand. The MVP will be deployed through the university's Careers and Employability Service, who will act as the organisational owner. Technical maintenance will sit with a small digital or innovation team, while ongoing content curation and case-study updates will involve collaboration between careers staff and academic colleagues.

Beneficiaries

The primary beneficiaries are domestic final-year students and recent graduates seeking employment in data and technology roles. These students benefit from clearer direction, reduced job-search overwhelm, and more targeted preparation. Secondary beneficiaries include the Careers and Employability Service, which gains a more scalable and data-informed tool, and academic schools, which may see improved Graduate Outcomes Survey results. Over time, employers may also benefit indirectly as better-prepared candidates enter the recruitment pipeline (CBI, 2022).

Measuring Success

Table 2 below summarises the proposed key success metrics for the pilot phase.

Success Metric	Target Outcome	Measurement Approach
User engagement	50+ students onboarded in pilot	Platform login and intake completion data
Action completion	>60% complete at least 2 of 3 weekly actions	In-app tracking and self-report
Perceived clarity	Increased confidence score post-use	Pre/post survey (Likert scale)
Return usage	>40% return for a second weekly plan	Session analytics
Feedback quality	Useful signal on which actions work	Post-action feedback form

Table 2: Proposed success metrics and measurement approaches for the LaunchPad Nexus MVP pilot. Source: Authors' own.

Next Steps

The immediate next steps following approval are: (1) confirm university sponsorship and Gemini API access with the digital infrastructure team; (2) recruit 50 pilot participants from the current final-year cohort in partnership with the Careers and Employability Service; (3) conduct a two-week internal testing phase before the full pilot launch; and (4) establish a review point at week 9 to assess pilot data and decide whether to

proceed to wider rollout. All student data will be anonymised, participation will be voluntary, and the tool will comply fully with UK GDPR (ICO, 2023).

Limitations

The MVP faces several challenges. Recommendation quality depends on accurate self-reported profile information, and LLM hallucinations may produce plausible but inaccurate advice, requiring human oversight before deployment. Data privacy requirements under the UK GDPR (ICO, 2023) mean all student inputs must be anonymised, and LLM API costs will scale with usage, requiring university subsidisation during the pilot. The tool also cannot resolve structural employment barriers such as discrimination or regional labour market conditions.

Conclusions and Future Scope

LaunchPad Nexus addresses a genuine and well-evidenced gap in graduate employability support: the absence of personalised, timely and action-focused guidance at the moment students review a specific job description. By using an LLM to filter noise from job descriptions and match actionable steps to the student's individual profile, the tool has the potential to meaningfully improve job-search clarity, confidence and targeting.

For long-term impact, the pilot data will be used to validate and improve the recommendation engine. Future enhancements could include AI-generated role-specific assessments, gamified progress tracking with streaks and milestones, a recruiter follow-up assistant, and peer learning features connecting students targeting similar roles. At scale, the platform could be offered as a white-label institutional dashboard (B2B) to other universities, generating a sustainable revenue stream while broadening impact.

References

Crawford, P. and Fink, W. (2019–2020) 'Employability Skills and Students Critical Growth Areas', NACTA Journal, Vol. 64 (Nov 2019–Oct 2020), pp. 132–141. North American Colleges and Teachers of Agriculture (NACTA).

Jackson, D. (2014) 'Factors influencing job attainment in recent Bachelor graduates: evidence from Australia', Higher Education, Vol. 68, No. 1 (July 2014), pp. 135–153. Springer.

James, S., Warhurst, C., Tholen, G. and Commander, J. (2013) 'What we know and what we need to know about graduate skills', Work, Employment & Society, Vol. 27, No. 6 (December 2013), pp. 952–963. Sage Publications, Ltd.

Robinson, J.S., Garton, B.L. and Vaughn, P.R. (2007) 'Becoming Employable: A Look at Graduates' and Supervisors' Perceptions of the Skills Needed for Employability', NACTA Journal, Vol. 51, No. 2 (June 2007), pp. 19–26. North American Colleges and Teachers of Agriculture (NACTA).

Smith, J., McKnight, A. and Naylor, R. (2000) 'Graduate Employability: Policy and Performance in Higher Education in the UK', The Economic Journal, Vol. 110, No. 464 (June 2000), pp. F382–F411. Oxford University Press on behalf of the Royal Economic Society.

Wadsworth, J. (2010) 'The UK Labour Market and Immigration', National Institute Economic Review, No. 213 (July 2010), pp. R35–R42. Cambridge University Press.

Wilton, N. (2011) 'Do employability skills really matter in the UK graduate labour market? The case of business and management graduates', Work, Employment & Society, Vol. 25, No. 1 (March 2011), pp. 85–100. Sage Publications, Ltd.

AGCAS (2023) Graduate labour market trends. Association of Graduate Careers Advisory Services. Available at: https://www.agcas.org.uk (Accessed: 10 March 2025).

CBI (2022) Employer perspectives on graduate recruitment. Confederation of British Industry. Available at: https://www.cbi.org.uk (Accessed: 12 March 2025).

HESA (2023) Graduate activities by provider and subject area of degree (Table 28). Higher Education Statistics Agency. Available at: https://www.hesa.ac.uk/data-and-analysis/graduates/table-28 (Accessed: 10 March 2025).

HESA (2024) Graduate outcomes: What do graduates do? Higher Education Statistics Agency. Available at: https://www.hesa.ac.uk/data-and-analysis/graduates (Accessed: 8 March 2025).

ICO (2023) Guide to the UK GDPR. Information Commissioner's Office. Available at: https://ico.org.uk/for-organisations/guide-to-data-protection/guide-to-the-uk-gdpr/ (Accessed: 15 March 2025).

ISE (2023) Student development survey 2023. Institute of Student Employers. Available at: https://ise.org.uk/research (Accessed: 11 March 2025).

Prospects (2023) Graduate jobs, careers advice and graduate schemes. Available at: https://www.prospects.ac.uk (Accessed: 9 March 2025).

Shury, J., Vivian, D., Turner, C. and Downing, C. (2017) Planning for success: Graduates' career planning and its effect on graduate outcomes. Department for Education. Available at: https://www.gov.uk/government/publications (Accessed: 10 March 2025).

Universities UK (2023) Universities improving graduate employment: Case studies. Available at: https://www.universitiesuk.ac.uk (Accessed: 9 March 2025).

Appendix

Appendix A: Empathy Map Canvas

Figure 1 below presents the Empathy Map developed for our primary customer segment: domestic final-year and recent graduate students seeking employment in data and technology roles. The map was constructed using insights gathered from informal peer interviews, student survey responses and observation of job-search behaviours within our own group. It informed the identification of key pain points and gains that shaped the design of LaunchPad Nexus.

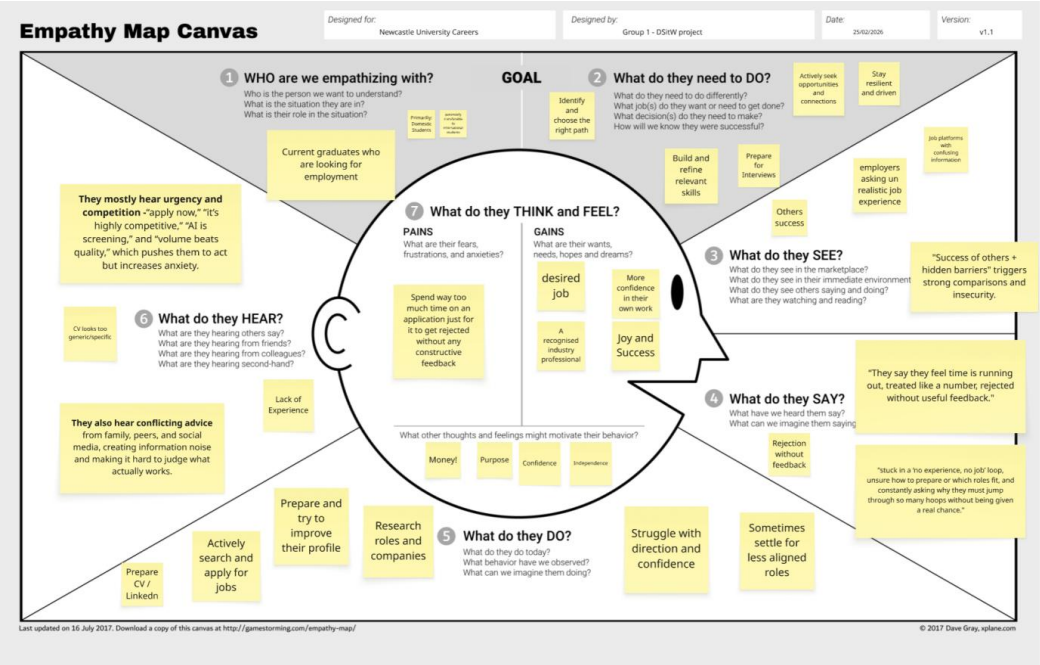


Figure 1: Empathy Map Canvas for the LaunchPad Nexus target customer segment — domestic final-year and recent graduate students seeking employment in data/technology roles.

The Empathy Map highlights that students think and feel a great deal of anxiety about whether their current actions are the right ones, and whether they are competitive enough. They hear conflicting advice from peers, tutors and social media, adding to confusion. They see an overwhelming number of job listings and requirements. They say they are unsure where to start. They do, in practice, often apply to large numbers of roles without a targeted strategy. Key pains include overwhelm, low response rates, and lack of personalised feedback. Key gains desired include a clear action plan, confidence in their applications, and a sense of progress and momentum.

Appendix B: Customer Journey Map

Figure 2 below presents the Customer Journey Map, which traces the experience of a domestic final-year student seeking employment from initial awareness of the job market through to securing an offer. The map was developed collaboratively by the group using a structured template and refined in response to feedback received during the module.

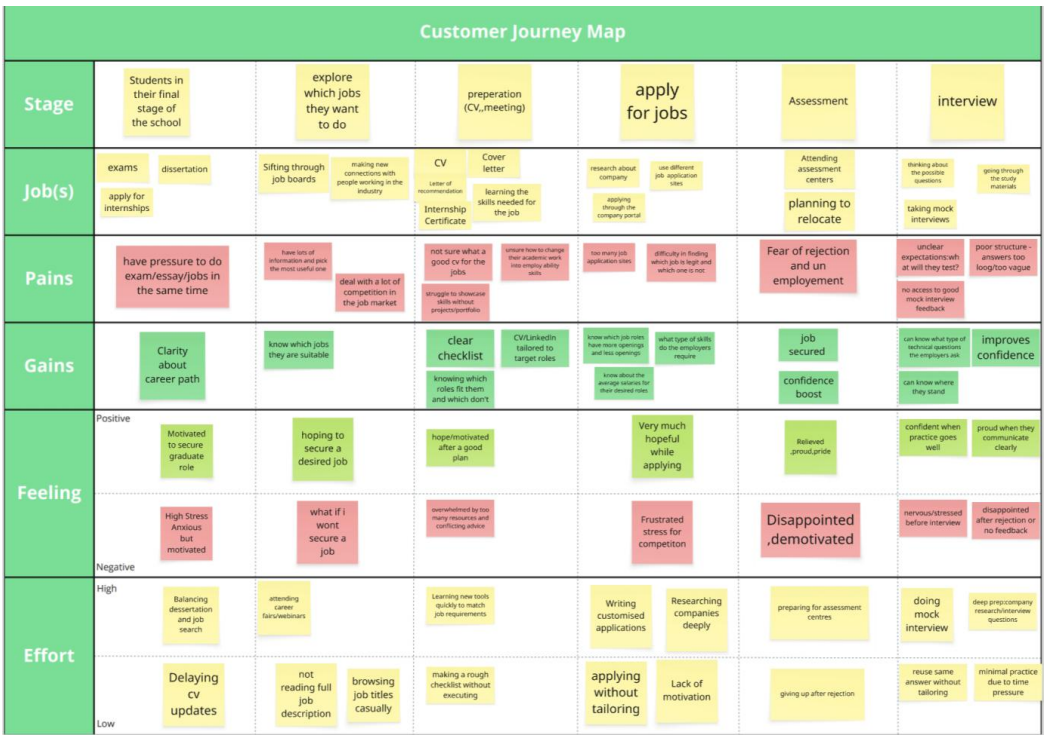


Figure 2: Customer Journey Map for a domestic final-year student navigating the graduate job search process, with key pain points, gains and emotional states at each stage.

The journey map identifies six key stages: Awareness (discovering the graduate job market); Exploration (researching roles and requirements, often feeling overwhelmed); Application Preparation (attempting to tailor CVs without clear direction); Submission (applying to multiple roles with varying targeting); Waiting and Response (a high-anxiety period with little feedback); and Outcome (progressing to interview or receiving rejection with minimal learning). Key pain points are concentrated in the Exploration and Application Preparation stages, directly informing the design of LaunchPad Nexus as a tool focused on the period between seeing a job description and submitting a strong application.

Appendix C: Value Proposition Canvas

Figure 3 below presents the completed Value Proposition Canvas for LaunchPad Nexus, developed using the standard template provided for the module. The left square (Value Map) sets out the Products & Services, Gain Creators and Pain Relievers that LaunchPad Nexus offers. The right circle (Customer Profile) sets out the Customer Jobs, Gains and Pains of our target segment: domestic final-year students and recent graduates seeking employment in data and technology roles. The canvas was used to validate that our proposed solution genuinely addresses the needs identified through the empathy map and customer journey analysis.

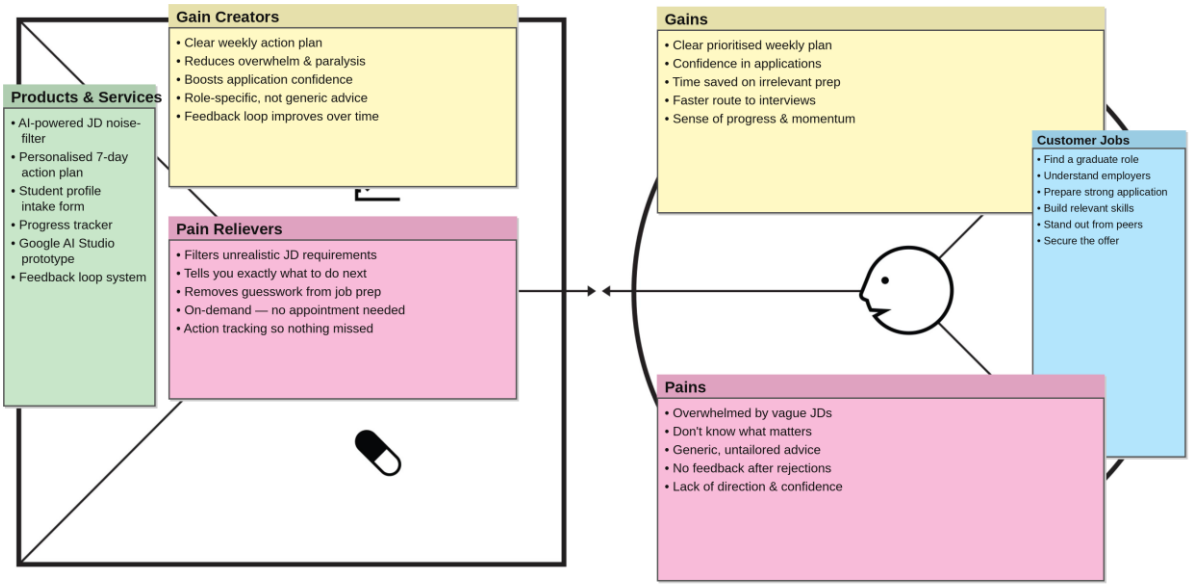


Figure 3: Completed Value Proposition Canvas for LaunchPad Nexus, showing the fit between the value map (left) and customer profile (right).

The canvas shows a strong fit between the value map and the customer profile. The noise-filtering feature directly relieves the pain of overwhelming job descriptions. The personalised weekly action plan relieves the pain of not knowing what to do next and directly creates the gain of a clear, prioritised plan. The feedback loop creates the gain of progress and momentum over time. Products and services such as the student profile intake form and the Google AI Studio prototype address the core customer job of preparing a strong, targeted application.

Appendix D: Business Model Canvas

Figure 4 below presents the Business Model Canvas for LaunchPad Nexus. The canvas was developed to articulate the key building blocks of the solution and its path to sustainable delivery, informing how the MVP would be resourced, delivered and scaled beyond the initial pilot.

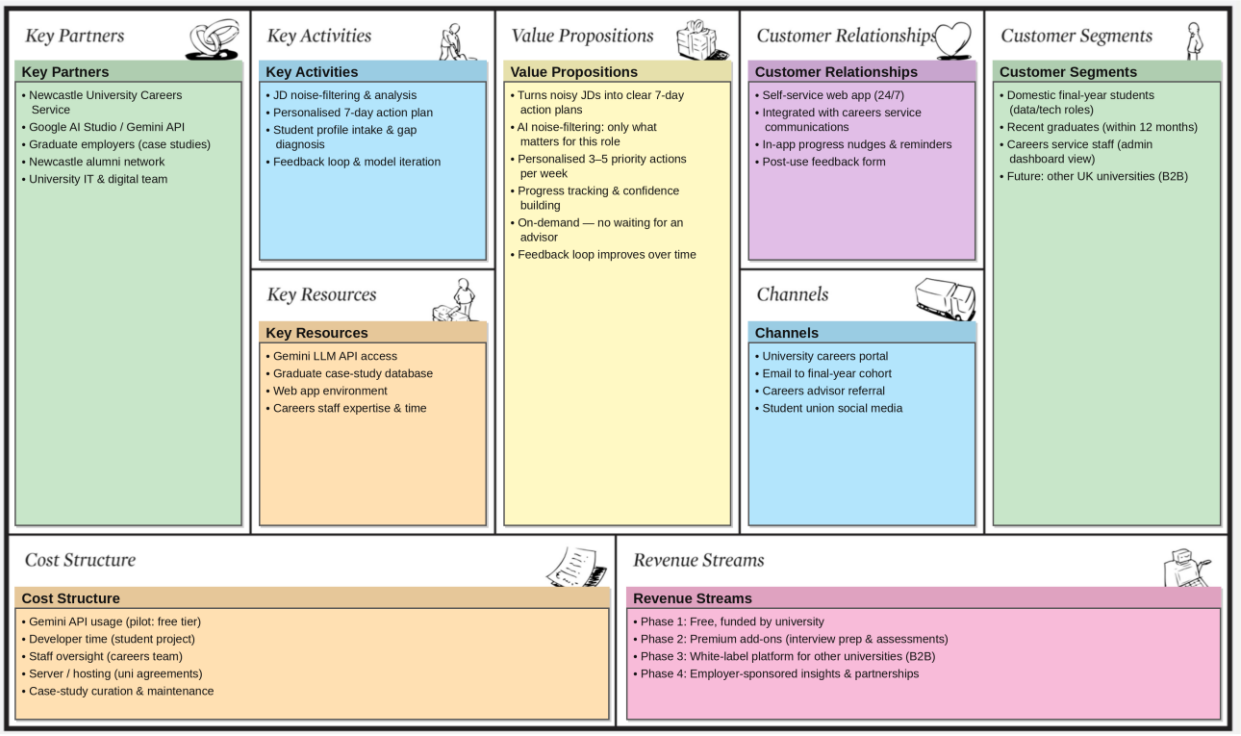


Figure 4: Completed Business Model Canvas for LaunchPad Nexus, showing key building blocks for MVP delivery and future scaling.

The Business Model Canvas highlights that the MVP can be delivered at low cost during the pilot phase, with the university's innovation budget providing seed support. Key partnerships include the Careers and Employability Service, Google AI Studio / Gemini API, and graduate employers who contribute case studies. The Value Proposition centres on turning noisy job descriptions into clear, personalised 7-day action plans. Long-term sustainability depends on demonstrating pilot value and then expanding to premium features (Phase 2) and institutional white-label licensing (Phase 3).

Appendix E: MVP Implementation Plan

Table 3 below outlines the phased implementation plan for the LaunchPad Nexus MVP, spanning approximately twelve weeks from initial setup to pilot evaluation.

Phase	Timeline	Key Activities	Owner / Resource
1: Setup & Partnerships	Weeks 1–2	Confirm university support; establish LLM API access; develop initial case-study database; build intake form prototype.	Digital team + Careers Service
2: Build & Test	Weeks 3–5	Develop backend recommendation engine; integrate LLM with case-study matching; build output action plan view; internal testing with 5–10 students.	Dev team + test cohort
3: Pilot Launch	Weeks 6–9	Recruit 50+ pilot users from final-year cohort; students use tool weekly; collect in-app feedback; monitor engagement metrics.	Careers Service + pilot cohort
4: Evaluate & Refine	Weeks 10–12	Analyse engagement data and feedback; assess action completion rates; conduct brief interviews with sample users; produce evaluation report.	Research/data team + Careers staff

Table 3: LaunchPad Nexus MVP implementation plan showing phased activities, timeline and ownership.

Key assumptions to be tested during the pilot include whether students are willing to input accurate profile information, whether the LLM recommendations are perceived as relevant and actionable, and whether completing the suggested actions improves students' confidence and job-search outcomes. Feedback collected during the pilot will be used to improve the recommendation engine and refine the case-study database ahead of any wider rollout. All student data will be anonymised, participation will be voluntary, and the tool will comply fully with the university's data protection policies and the UK GDPR (ICO, 2023).

Appendix F: Prototype Demo — LaunchPad Nexus Web App

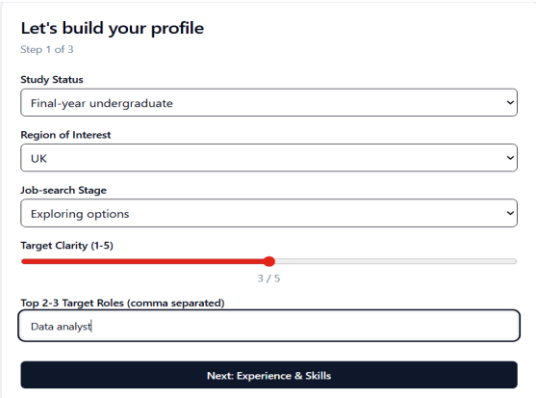
As part of this module, our group developed and demonstrated a working prototype of LaunchPad Nexus, built as a web application using Google AI Studio and Firebase. The prototype is accessible at the following URL:

`https://ai.studio/apps/4d90c155-d85e-4e87-9d06-9dc16a439c8d`

The prototype is a structured web application powered by Google's Gemini large language model, with Firebase used as the backend database. Unlike a simple chatbot, it presents students with a clean, multi-step form interface that guides them through profile building, job description analysis, and action plan generation. The application was demonstrated live during the group presentation on 13th March 2025.

Step 1 — Profile Builder (3-Step Wizard)

The student completes a structured three-step profile wizard. Step 1 (Figure 6) collects their Study Status (e.g. Final-year undergraduate), Region of Interest (e.g. UK), Job-search Stage (e.g. Exploring options), a Target Clarity rating from 1 to 5, and their top 2–3 target roles. Step 2 (Figure 7) collects experience indicators (internship, significant project, hackathon, open-source contribution) and self-rated skill confidence scores from 1 to 5 across Programming, SQL, Data Visualisation, Statistics, and Communication.



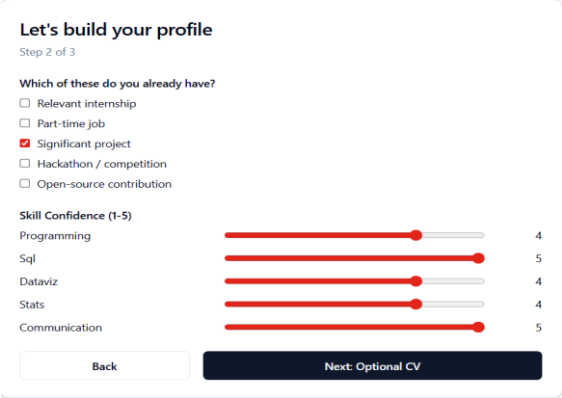


Figure 6 (left): Profile Builder — Step 1 of 3, collecting study status, region, job-search stage and target roles. Figure 7 (right): Profile Builder — Step 2 of 3, collecting experience and skill confidence ratings.

Step 2 — Job Description Analysis

Once the profile is complete, the student is taken to the "Analyse a Job Description" screen (Figure 8). They enter the Role Title, an optional Company Name, and paste the full job description text into a large text area. This structured input allows the Gemini model to contextualise the analysis against both the JD content and the student's profile.

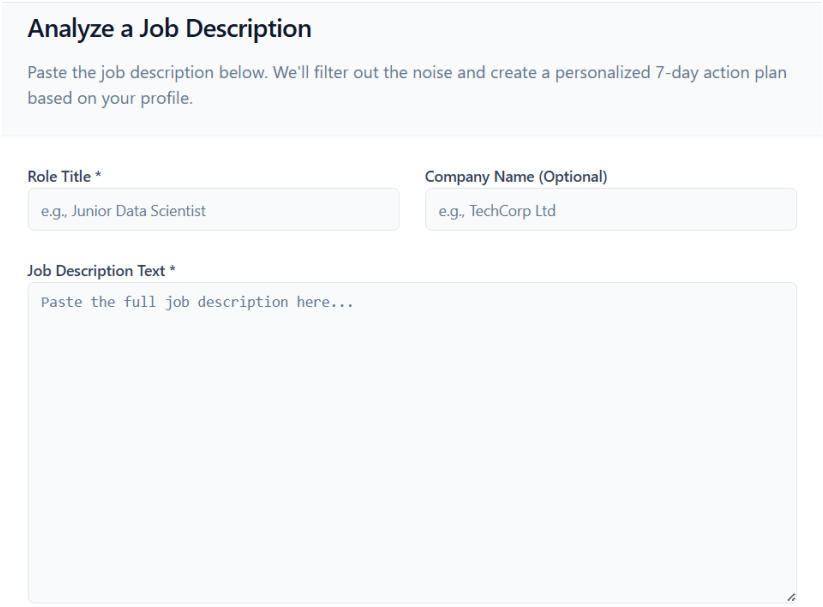


Figure 8: Job Description input screen — students enter the role title, company name and paste the full JD text for analysis.

Step 3 — Output: De-Noised JD and 7-Day Action Plan

The system then generates a rich, structured output (Figures 9 and 10) divided into several sections. The "Role & Company at a Glance" panel provides a concise contextual summary of the employer and role. The "Job Description De-Noised" section identifies whether the JD is realistic or a 'unicorn hunt' (asking for skills beyond the stated level), and clearly separates "What You Must Focus On" from "What You Should Ignore For Now". A "Red Flags & Notes" panel highlights issues such as title mismatches, vague requirements, or missing salary information. Finally, a full "Your 7-Day Action Plan" is generated with numbered steps, each containing an estimated time, an explanation of why the action matters, and 3–4 specific sub-steps with direct links to relevant resources such as datasets, tutorials, or documentation.

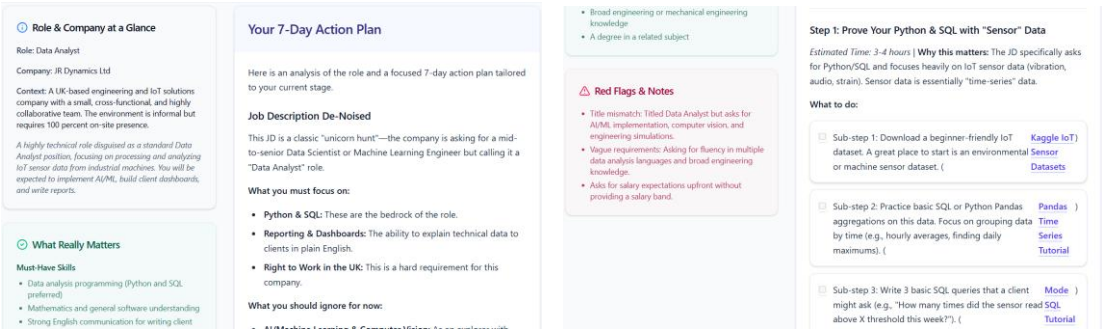


Figure 9 (left): Output showing Role & Company at a Glance, What Really Matters, Red Flags & Notes, and the start of the 7-Day Action Plan. Figure 10 (right): A detailed action step with estimated time, rationale, and sub-steps with resource links.

Technology Stack

The prototype was built using Google AI Studio's app-building interface, with a custom system prompt configuring the Gemini model's analysis behaviour. Firebase was used as the backend database, enabling persistent storage of student profiles and session data. The front end is a structured web application with a

clean multi-step form UI rather than a freeform chat interface. This approach validated the core concept rapidly within the module timeline. In a production version, the system prompt logic would be replaced by a more robust backend pipeline integrating a curated case-study database, user authentication, persistent action tracking, and a feedback collection mechanism as described in Appendix E.

Prototype Limitations

The prototype served as a proof-of-concept and carries several acknowledged limitations. The skill confidence sliders and experience checkboxes in the profile builder are self-reported and not validated, meaning the quality of recommendations depends on student honesty and self-awareness. The Gemini model is subject to occasional hallucinations and may produce confident but inaccurate advice, particularly for niche sectors or highly specialised roles. Firebase currently stores profiles per session without user authentication, so data is not linked across sessions. These limitations would all be addressed in the full MVP through the structured data pipeline, case-study matching layer, user authentication, and human oversight mechanisms described in Appendix E.

Despite these limitations, the prototype successfully demonstrated the core value proposition of LaunchPad Nexus: that an AI-powered web tool can take a noisy, overwhelming job description, combine it with a structured student profile, and return a clear, de-noised analysis and actionable 7-day plan within seconds. The output quality — particularly the 'unicorn hunt' detection, the Red Flags panel, and the sub-step resource links — was noted positively by the presentation audience as practically useful and significantly above the quality of generic careers advice.

Appendix G: Initial Survey — LaunchPad Nexus Job Search Support

As part of the early exploratory work for this project, the group designed and distributed a short survey to understand the job-search experiences and needs of the target customer segment: final-year and recent graduate students. The survey was built and distributed via Google Forms and is accessible at the following URL:

`https://forms.gle/tFB34CPSQ5Pxeg8r5`

The survey was titled "LaunchPad Nexus – Job Search Support (Prototype)" and described its purpose to participants as follows: "We're testing a student-built idea to help graduates and postgraduates get into the right jobs, faster. This form asks about your situation and then suggests a small set of next steps. It's a prototype, so please be honest and ignore any rough edges."

The survey served two purposes simultaneously: first, as a primary research instrument to understand the target user's current situation, motivations, pain points and preferred areas of focus; and second, as an early-stage prototype of the intake form itself, allowing the group to test whether students were willing to provide the kind of structured profile information that LaunchPad Nexus would require in its full MVP. This dual-purpose design was intentional and informed both the customer journey mapping and the MVP specification.

Survey Questions

The full survey contained the following questions, organised across multiple pages. All questions were required unless otherwise stated.

Page 1: About You

Question 1. What best describes you right now?

- Final-year undergraduate
- Final-year postgraduate (taught MSc/MA)
- Recent graduate (within 1 year)
- Other (open text)

Question 2. Are you an international or domestic student?

- International
- Domestic

Question 3. Which broad area are you mainly targeting for jobs?

- Data / AI / tech roles
- Business / consulting / finance
- Other professional roles
- I'm not sure yet

Question 4. Where are you mainly hoping to work?

- UK
- My home country
- Open to multiple countries

- Not sure yet

Page 2: Your Job Search

Question 5. How would you describe your current job search stage?

- I haven't really started yet
- I'm researching roles and sectors
- I'm actively applying
- I've had some interviews but no offer yet
- I already have something lined up

Question 6. What is your biggest challenge right now when it comes to finding a job?

- I don't know which jobs I'm suited for
- Job descriptions feel overwhelming or unclear
- I'm not sure what skills I'm missing
- I'm applying but not getting responses
- I lack confidence in interviews
- I'm struggling to balance job-searching with studying
- Other (open text)

Question 7. When you look at a job description, how do you usually decide what to focus on to prepare?

- I try to cover everything listed
- I focus on the requirements marked as 'essential'
- I compare it to my CV and identify gaps
- I ask someone (friend, tutor, careers advisor)
- I search online for advice
- I'm not sure — it feels overwhelming

Page 3: What Would Help

Question 8. How useful would it be to have a tool that reads a job description and tells you exactly what to focus on for the next 7 days?

- Very useful — I'd use it immediately
- Quite useful — I'd try it
- Might be useful — depends on how good the advice is
- Not sure
- Not useful — I prefer figuring it out myself

Question 9. What would make you most likely to trust and act on AI-generated job search advice? (Tick all that apply)

- If it was based on real graduate case studies
- If it came via the university careers service
- If other students had found it helpful
- If I could see why each action was recommended
- If I could give feedback and the advice improved

- Nothing — I wouldn't trust it

Question 10. How many minutes per week would you realistically spend using a job-search support tool like this?

- Less than 10 minutes
- 10–20 minutes
- 20–40 minutes
- More than 40 minutes

Page 4: About You (Profile)

Question 11. What degree subject or field are you studying / did you study?

(Open text field)

Question 12. What is your target job title or role type? (e.g. Data Analyst, Software Engineer, Business Analyst)

(Open text field)

Question 13. How confident do you feel about your job search right now? (1 = not at all confident, 5 = very confident)

- 1 — Not at all confident
- 2
- 3 — Somewhat confident
- 4
- 5 — Very confident

Question 14. Is there anything else you'd like us to know about your job search situation or what you'd find helpful?

(Optional open text field)

Purpose and Use of Survey Data

The survey responses informed two key areas of the project. First, Questions 1–4 (demographic and contextual questions) helped validate our decision to focus on domestic final-year and recent graduate students targeting data and technology roles. Second, Questions 5–10 (job search experience and needs) provided the primary evidence base for the customer journey map (Appendix B) and the empathy map (Appendix A), confirming that the most significant pain point was not a lack of job listings but an inability to know what to focus on when reviewing a specific job description. Third, Questions 11–14 (profile questions) were directly used to design the student intake form in the prototype, testing whether students were willing to provide the structured information that the LaunchPad Nexus recommendation engine would need.

The survey was distributed informally within the student cohort during the early stages of the module and was not intended as a statistically representative sample. Its purpose was primarily exploratory and formative, generating qualitative insight and directional evidence rather than definitive quantitative findings. The responses nonetheless provided strong directional support for all four sections of the report: Context, Need, Vision and Outcomes.

Survey Results

The survey collected six responses from the target cohort. Although the sample size is small and not statistically representative, the results offered strong directional evidence that validated the core assumptions underlying LaunchPad Nexus. The three charts below summarise the most relevant findings.

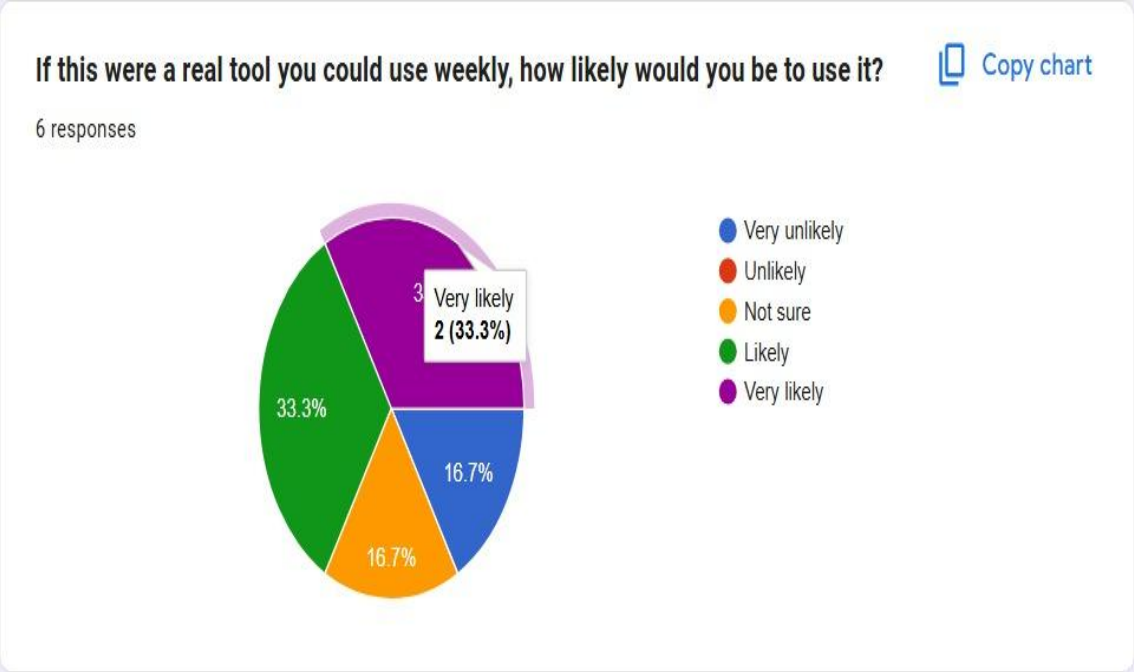


Figure 11: Pain Points — What feels hardest about job searching right now? (Q8, 6 responses, select up to 3). "Knowing where to start" was the top pain point (50%), followed by six further challenges each cited by 33.3% of respondents.

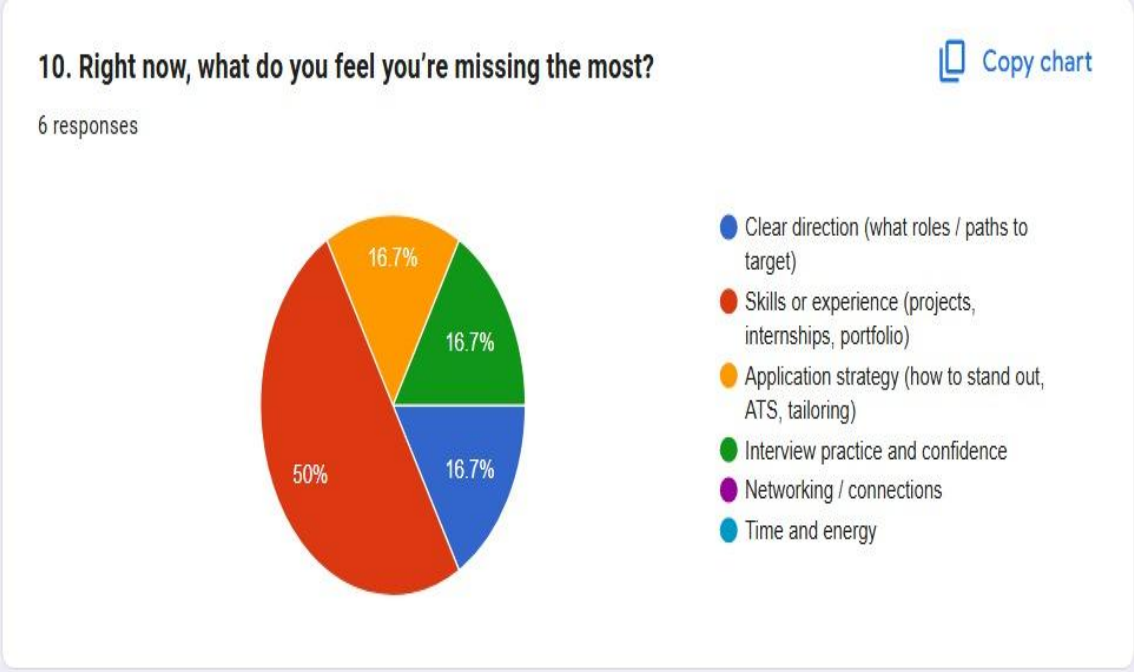


Figure 12: Weekly Usage Likelihood — How likely would you be to use this tool weekly? (6 responses). 66.6% of respondents said "Likely" or "Very likely", with 16.7% "Not sure" and 16.7% "Very unlikely".

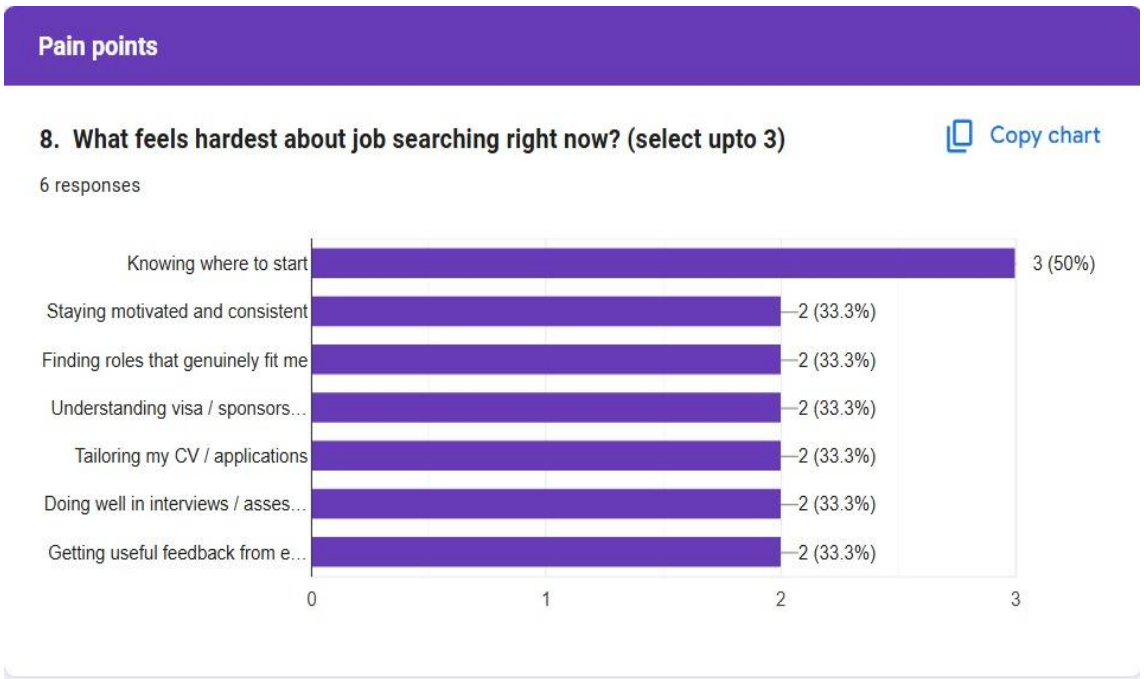


Figure 13: What are you missing most? (Q10, 6 responses). Half of respondents (50%) identified skills or experience as their most pressing gap. Clear direction, application strategy, and interview practice each accounted for 16.7%.

Discussion of Survey Findings

The survey only captured six responses, so we're not claiming it's definitive — but what came back was consistent and pointed clearly in the right direction.

The pain-points question (Figure 11) was the most telling. Half of respondents said their biggest struggle was simply knowing where to start. On top of that, a third each flagged staying motivated, finding roles that fit, CV tailoring, interview prep, and getting useful feedback. The picture that emerges isn't a student with one specific gap — it's someone who feels lost across the whole process. That's exactly what LaunchPad Nexus is designed to cut through.

The usage data (Figure 12) was more encouraging than we expected. Two thirds said they'd be likely or very likely to use a tool like this every week — notably high for something still at prototype stage. The small number who were unsure are probably just earlier in their job search and haven't hit the wall yet.

Figure 13 was the one that most shaped our thinking about where to take the product next. Half of respondents pointed to a lack of real-world experience — projects, internships, portfolio work. That told us that telling students what to do isn't enough; they also need help building the evidence employers actually want to see, which is something we want to tackle in a future version.

Taken together, the survey gave us enough to feel confident we were solving a real problem, for real people, in a way they would actually want to use.